

Competency

To verify experimentally and prove theoretically the properties of triangles

Learning Outcomes

- To differentiate between the theoretical proof and experimental verification on properties of triangles
- To prove the theorems on triangles and verify the statements by inductive method (theorems which cannot be proved theoretically)
- To prove the theorems based on mid-point theorem

Level-wise learning objectives

S.N.	LEVELS	OBJECTIVES
1.	Knowledge (K)	- To identify centroid and orthocentre of triangle - To recall the sum of interior angles of triangle - To tell the relation of exterior angle and the opposite exterior angles of triangle - To relate the sum of any two sides and the third side of the triangle - To state the mid-point theorem
2.	Understanding (U)	- To find the unknown angles using the properties of triangle - To find the unknown length of sides of angles of triangles by using mid-point theorem
3.	Application (A)	- To verify the properties of triangle experimentally - To prove the theorems based on properties of triangle
4.	High Ability (HA)	- To prove the unseen theorems based on properties of triangle

Required Teaching Materials/ Resources

Geo board, rubber-bands, mechano strip, sticks, colourful chart paper, models of triangles, rectangular sheet of paper, ICT tools etc.

Pre-knowledge: Types of triangles, basic properties of triangles

Teaching Activities

1. For warm-up, discuss upon the types of triangles based on sides or angles with models of triangles on chart papers or geo-board or mechano strips or sticks
2. Show the centroid and orthocentre of the triangles by paper folding.
3. Discuss the following properties of triangles experimentally and theoretically
 - The sum of angles of triangle is equal to two right angles
 - The exterior angle of a triangle is equal to the sum of the two opposite interior angles
 - The sum of any two sides of a triangle is greater than the third side
 - In any triangle, the angle opposite to the longer side is greater than the angle opposite to the shorter side
 - In any triangle, the side opposite to the greater angle is longer than the side opposite to the smaller angle

- Of all straight line segments drawn to the given line from a point outside it, the perpendicular is the shortest one.
 - Base angles of isosceles triangle are equal
 - if two angles of triangle are equal, it is an isosceles triangle
 - The bisector of vertical angle of an isosceles triangle is perpendicular bisector of the base.
 - The straight line joining the vertex and the mid-point of the base of an isosceles triangle is perpendicular to the base and bisects the vertical angle.
 - The straight line joining the mid-points of any two sides of a triangle is parallel to the third side and it is equal to half of the length of the third side.
- The straight line segment drawn through the mid-point of one side of a triangle and parallel to the another side bisects the third side
 - Divide the students into groups and give the theorems to prove them theoretically.
 - Focus on project work

Solution of selected problems from Vedanta Excel in Mathematics

A. Angles of Triangle.

1. In $\triangle ABC$, $2\angle A = 3\angle B = 6\angle C$, find $\angle A$, $\angle B$ and $\angle C$.

Solution: Here, $2\angle A = 3\angle B = 6\angle C$

$$\therefore \angle A = 3\angle C \text{ and } \angle B = 2\angle C$$

Now, $\angle A + \angle B + \angle C = 180^\circ$ [$\because$ Being the sum of angle of Δ]

$$\text{or, } 3\angle C + 2\angle C + \angle C = 180^\circ$$

$$\text{or, } 6\angle C = 180$$

$$\therefore \angle C = 30^\circ$$

Also, $\angle A = 3\angle C = 3 \times 30^\circ = 90^\circ$ and $\angle B = 2\angle C = 2 \times 30^\circ = 60^\circ$

Hence, $\angle A = 90^\circ$, $\angle B = 60^\circ$ and $\angle C = 30^\circ$

2. In $\triangle PQR$, $\angle P - \angle Q = 20^\circ$ and $\angle Q - \angle R = 50^\circ$, find $\angle P$, $\angle Q$ and $\angle R$.

Solution: Here, In $\triangle PQR$, $\angle P - \angle Q = 20^\circ$ (i)

$$\angle Q - \angle R = 50^\circ \quad \text{..... (ii)}$$

$$\text{But, } \angle P + \angle Q + \angle R = 180^\circ \quad \text{..... (iii)}$$

Now, adding equations (i), (ii) and (iii) we get

$$(\angle P - \angle Q) + (\angle Q - \angle R) + (\angle P + \angle Q + \angle R) = 20^\circ + 50^\circ + 180^\circ$$

$$\text{or, } 2\angle P + \angle Q = 250 \quad \text{..... (iv)}$$

Again adding equation (i) and (iv), we get

$$(\angle P - \angle Q) + (2\angle P + \angle Q) = 20^\circ + 250^\circ$$

$$\text{or, } 3\angle P = 270^\circ \quad \therefore \angle P = 90^\circ$$

Substituting the value of $\angle P$ in equation (i), we get

$$90^\circ - \angle Q = 20^\circ \quad \therefore \angle Q = 70^\circ$$

Substituting the value of $\angle Q$ in equation (ii), we get

$$70^\circ - \angle R = 50^\circ \quad \therefore \angle R = 20^\circ$$

Hence, $\angle P = 90^\circ$, $\angle Q = 70^\circ$ and $\angle R = 20^\circ$.

3. Find the unknown sizes of angles in the given figure.

Solution:

$$\text{i) In } \triangle ABC, \angle A + \angle B + \angle C$$

$$= 180^\circ$$

$$\text{or, } (x + x) + 65^\circ + 35^\circ$$

$$= 180^\circ$$

$$\text{or, } 2x$$

$$= 80^\circ \text{ or, } \therefore x = 40^\circ$$

$$\text{ii) In } \triangle ABD; \angle BAD + \angle ABD + \angle ADB$$

$$= 180^\circ$$

$$\text{or, } x + 65^\circ + y$$

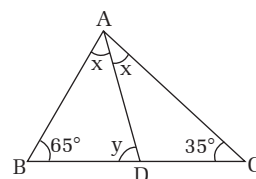
$$= 180^\circ$$

$$\text{or, } 40^\circ + 65^\circ + y$$

$$= 180$$

$$\text{or, } \therefore y = 75^\circ$$

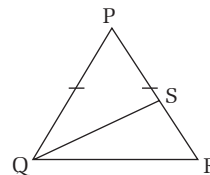
Hence, $x = 40^\circ$ and $y = 75^\circ$



4. In the given figure, $PQ = PR$ and QS is the bisector of $\angle PQR$. If $\angle PQS = 30^\circ$, find $\angle PSQ$.

Solution:

- i) $\angle PQS = \angle SQR = 30^\circ$ [As QS the bisector of $\angle PQR$]
 ii) $\angle PQR = \angle PRQ = 60^\circ$ [$\because PQ = PR$]
 iii) In $\triangle PQR$; $\angle PQR + \angle PRS + \angle QPR = 180^\circ$
 or, $60^\circ + 60^\circ + \angle QPR = 180^\circ$
 $\therefore \angle QPR = 60^\circ$
 iv) In $\triangle PQS$; $\angle PQS + \angle PSQ + \angle QPS = 180^\circ$
 or, $30 + \angle PSQ + 60^\circ = 180^\circ$
 $\therefore \angle PSQ = 90^\circ$



5. In the adjoining figure, the bisectors of $\angle ABC$ and $\angle ACB$ of $\triangle ABC$ meet at O . Prove that $\angle BOC = 90^\circ + \frac{\angle A}{2}$.

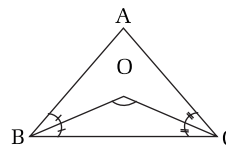
Solution:

Here,

Given: In $\triangle ABC$, the bisectors of $\angle ABC$ and $\angle ACB$ meet at O .

To prove: $\angle BOC = 90^\circ + \frac{\angle A}{2}$

Proof:



	Statements		Reasons
1.	$\angle ABC = 2\angle OBC$ and $\angle ACB = 2\angle OCB$	1.	Given
2.	In $\triangle ABC$; $\angle ABC + \angle ACB + \angle BAC = 180^\circ$ or, $2\angle OBC + 2\angle OCB + \angle BAC = 180^\circ$ or, $2(\angle OBC + \angle OCB) = 180^\circ - \angle BAC$ $\therefore \angle OBC + \angle OCB = \frac{180^\circ - \angle A}{2} = 90^\circ - \frac{\angle A}{2}$	2.	
3.	In $\triangle BOC$; $\angle BOC + \angle OBC + \angle OCB = 180^\circ$ or, $\angle BOC + 90^\circ - \frac{\angle A}{2} = 180^\circ$ $\therefore \angle BOC = 180^\circ - 90^\circ + \frac{\angle A}{2}$ $= 90^\circ + \frac{\angle A}{2}$	3.	From (ii)

Proved

6. In the figure alongside, BP and CP are the angular bisectors of the exterior angles $\angle CBP$ and $\angle BCP$ of $\triangle ABC$. Prove that $\angle BPC = 90^\circ - \frac{\angle A}{2}$.

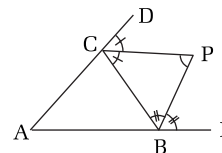
Solution:

Here,

Given: In $\triangle ABC$; BP and CP are the angular bisector of $\angle CBE$ and $\angle BCD$ respectively.
 $\therefore \angle CBE = 2\angle CBP$ and $\angle BCD = 2\angle BCP$.

To prove: $\angle BPC = 90^\circ - \frac{\angle A}{2}$

Proof:



	Statements		Reasons
1.	$\angle CBE = 2\angle CBP$ and $\angle BCD = 2\angle BCP$	1.	Given
2.	$\angle ABC = 180^\circ - \angle CBE$ and $\angle ACB = 180^\circ - \angle BCD$	2.	Supplementary angles

3.	In $\triangle ABC$, $\angle ABC + \angle ACB + \angle BAC = 180^\circ$ or, $(180^\circ - \angle CBE) + (180^\circ - \angle BCD) + \angle BAC = 180^\circ$ or, $-\angle CBE - \angle BCD + \angle BAC + 180^\circ = 0$ $\therefore \angle BAC + 180^\circ = \angle CBE + \angle BCD$	3.	Sum of angles of triangle and from (2).
4.	$\angle A + 180^\circ = 2\angle CBP + 2\angle BCP$ $\therefore \angle CBP + \angle BCP = 90^\circ + \frac{\angle A}{2}$	4.	From (1) and (3)
5.	In $\triangle BCP$, $\angle CBP + \angle BCP + \angle BPC = 180^\circ$ or, $90^\circ + \frac{\angle A}{2} + \angle BPC = 180^\circ$ $\therefore \angle BPC = 90^\circ - \frac{\angle A}{2}$	5.	Sum of angles of triangle and from (4).

Proved

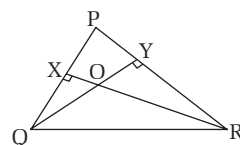
7. In $\triangle PQR$, $RX \perp PQ$ and $QY \perp PR$, RX and QY intersect at O .
Prove that $\angle QOR = 180^\circ - \angle P$.

Solution: Here,

Given: In $\triangle PQR$, $RX \perp PQ$ and $QY \perp PR$, RX and QY intersect at O .

To prove: $\angle QOR = 180^\circ - \angle P$.

Proof



	Statements		Reasons
1.	In $\triangle PQR$, $\angle PQR + \angle QRP + \angle QPR = 180^\circ$	1.	Sum of angles of triangle is 180°
2.	In $\triangle QXR$, $\angle XQR + \angle QRX + \angle QXR = 180^\circ$ or, $\angle XQR + \angle QRX + 90^\circ = 180^\circ$ $\therefore \angle XQR = 90^\circ - \angle QRX$	2.	Given
3.	In $\triangle QRY$, $\angle QRY + \angle RYQ + \angle YQR = 180^\circ$ or, $\angle QRY + 90^\circ + \angle YQR = 180^\circ$ $\therefore \angle QRY = 90^\circ - \angle YQR$	3.	
4.	$(90^\circ - \angle QRX) + (90^\circ - \angle YQR) + \angle P = 180^\circ$ or, $\angle P = \angle QRX + \angle YQR$	4.	From (1), (2) and (3)
5.	In $\triangle QOR$, $\angle QOR + \angle QRO + \angle ORQ = 180^\circ$ or, $\angle QOR + \angle P = 180^\circ$ $\therefore \angle QOR = 180^\circ - \angle P$	5.	From (4)

Proved

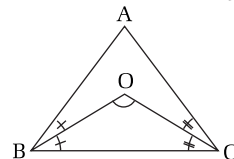
8. In the adjoining figure, ABC is an isosceles triangle. BO and CO are the bisectors of $\angle ABC$ and $\angle ACB$ respectively. Prove that BOC is an isosceles triangle.

Solution:

Given: In $\triangle ABC$, $AB = AC$, BO and CO are the bisectors of $\angle ABC$ and $\angle ACB$ respectively.

To prove: $\triangle BOC$ is also the isosceles triangle.

Proof



	Statements		Reasons
1.	$\angle ABC = \angle ACB$	1.	$AB = AC$
2.	$\angle ABC = 2\angle OBC$ and $\angle ACB = 2\angle OCB$	2.	Given
3.	$2\angle OBC = 2\angle OCB \therefore \angle OBC = \angle OCB$	3.	From (1) and (2)
4.	$\triangle BOC$ is an isosceles triangle.	4.	From (3), being base angles equal.

Proved

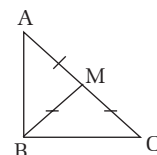
9. In the given figure, $AM = BM = CM$. Prove that $\triangle ABC$ is a right angled triangle.

Solution:

Given: $AM = BM = CM$

To prove: $\triangle ABC$ is a right angle triangle

Proof



	Statements		Reasons
1.	$\angle MAB = \angle MBA$	1.	$AM = BM$
2.	$\angle MBC = \angle MCB$	2.	$BM = CM$
3.	In $\triangle ABC$, $\angle ABC + \angle BCA + \angle BAC = 180^\circ$	3.	Sum of angles of triangle is 180°
4.	$(\angle MBA + \angle MBC) + \angle MCB + \angle MAB = 180^\circ$ or, $\angle MBA + \angle MBC + \angle MBC + \angle MBA = 180^\circ$ or, $2(\angle MBA + \angle MBC) = 180^\circ$ or, $\angle ABC = 90^\circ$	4.	From (1), (2) and (3) $\therefore \angle MBA + \angle MBC = \angle ABC$

Hence, $\triangle ABC$ is a right angle triangle.

10. In the given figure ABCD,

Prove that: $\angle BCD = \angle BAD + \angle ABC + \angle ADC$.

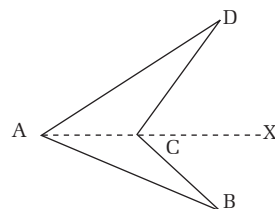
Solution:

Given: ABCD is an arrowhead.

To prove: $\angle BCD = \angle BAD + \angle ABC + \angle ADC$

Construction: AC is produced to X.

Proof



	Statements		Reasons
1.	In $\triangle ABC$; $\angle BCX = \angle CAB + \angle ABC$	1.	Being ext. angle of $\triangle$ equal to the sum opposite interior angles
2.	In $\triangle ACD$; $\angle DCX = \angle ADC + \angle CAD$	2.	Same as (1)
3.	$\angle BCX + \angle DCX = \angle CAB + \angle ABC + \angle ADC + \angle CAD$	3.	Adding (1) and (2)
4.	$\angle BCD = \angle BAD + \angle ABC + \angle ADC$	4.	By whole part axiom

Proved

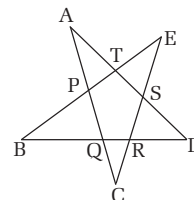
11. In the adjoining star shaped figure, prove that $\angle A + \angle B + \angle C + \angle D + \angle E = 180^\circ$

Solution:

Given: ABCDE is a star shaped figure.

To prove: $\angle A + \angle B + \angle C + \angle D + \angle E = 180^\circ$

Proof:



	Statements		Reasons
1.	In $\triangle ADQ$, $\angle A + \angle D = \angle DQC$	1.	Being ext. angle of $\triangle$ equal to the sum of opposite interior angles
2.	In $\triangle BRE$, $\angle B + \angle E = \angle BRC$	2.	Same as (1)
3.	In $\triangle QCR$, $\angle C + \angle DQC + \angle BRC = 180^\circ$	3.	Sum of angles of a triangle is 180°

4.	$\angle C + \angle A + \angle D + \angle B + \angle E = 180$ $\therefore \angle A + \angle B + \angle C + \angle D + \angle E = 180$	4.	From (1), (2) and (3)
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12. In the figure alongside, BE and CE are the angular bisector of $\angle ABC$ and $\angle ACD$ respectively. Prove that: $\angle BAC = 2\angle BEC$.

Solution:

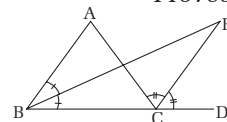
Given: BE and CE are the angular bisectors of $\angle ABC$ and $\angle ACD$ respectively.

To prove: $\angle BAC = 2\angle BEC$

Proof

	Statements		Reasons
1.	$\angle ABC = 2\angle EBC$ and $\angle ACD = 2\angle ECD$	1.	Given
2.	In $\triangle ABC$, $\angle BAC + \angle ABC = \angle ACD$	2.	Being ext. angle of $\triangle$ equal to sum of opposite interior angles
3.	$\angle BAC + 2\angle EBC = 2\angle ECD$ or, $\angle ECD = \frac{1}{2}(\angle BAC + 2\angle EBC)$	3.	From (1) and (2)
4.	In $\triangle BCE$; $\angle EBC + \angle BEC = \angle ECD$	4.	Same as (2)
5.	$\angle EBC + \angle BEC = \frac{1}{2}(\angle BAC + 2\angle EBC)$ or, $2\angle EBC + 2\angle BEC = \angle BAC + 2\angle EBC$ $\therefore \angle BAC = 2\angle BEC$	5.	From (3) and (4)

Proved



13. In the given figure, the bisector of $\angle ACU$ meets AU at O.

Prove that: $\angle COT = \frac{1}{2}(\angle CAT + \angle CUT)$

Solution:

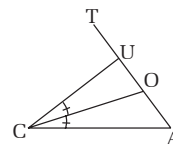
Given: In $\triangle CAU$, the bisector CO of $\angle ACU$ meets AT at O.

To prove: $\angle COT = \frac{1}{2}(\angle CAT + \angle CUT)$

Proof

	Statements		Reasons
1.	In $\triangle CAO$; $\angle COT = \angle OCA + \angle CAT$ $\therefore \angle OCA = \angle COT - \angle CAT$	1.	Being ext. angle $\triangle$ equal to the sum of opposite interior angles
2.	In $\triangle CAU$, $\angle CUT = \angle UCA + \angle CAT$	2.	Same as (1)
3.	$\angle UCA = 2\angle OCA$	3.	Given
4.	$\angle CUT = 2\angle OCA + \angle CAT$ $\therefore \angle OCA = \frac{1}{2}(\angle CUT - \angle CAT)$	4.	From (2) and (3)
5.	$\angle COT - \angle CAT = \frac{1}{2}(\angle CUT - \angle CAT)$ or, $2\angle COT - 2\angle CAT = \angle CUT - \angle CAT$ or, $2\angle COT = \angle CAT + \angle CUT$ $\therefore \angle COT = \frac{1}{2}(\angle CAT + \angle CUT)$	5.	From (1) and (4)

Proved



Proved

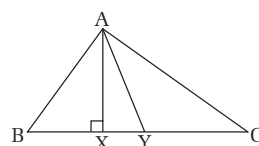
14. In the adjoining $\triangle ABC$, AY is the bisector of $\angle BAC$ and $AX \perp BC$. Prove that: $\angle XAY = \frac{1}{2}(\angle B - \angle C)$

Solution:

Given: In $\triangle ABC$, AY is the bisector of $\angle BAC$ and $AX \perp BC$.

To prove: $\angle XAY = \frac{1}{2}(\angle B - \angle C)$

Proof:



	Statements		Reasons
1.	$\angle BAC = 2\angle BAY$	1.	Given
2.	In $\triangle ABC$, $\angle BAC + \angle ABC + \angle ACB = 180^\circ$	2.	Sum of angles of $\triangle$ is 180°
3.	$2\angle BAY + \angle B + \angle C = 180^\circ$ $\therefore \angle BAY = \frac{1}{2}(180^\circ - \angle B - \angle C)$	3.	From (1) and (2)
4.	In $\triangle BAX$, $\angle BAX + \angle ABX + \angle AXB = 180^\circ$ or, $\angle BAX + \angle B + 90^\circ = 180^\circ$ $\therefore \angle BAX = 180^\circ - \angle B - 90^\circ = 90^\circ - \angle B$	4.	Sum of angles of $\triangle$ is 180°
5.	$\angle BAY - \angle BAX = \frac{1}{2}(180^\circ - \angle B - \angle C) - (90^\circ - \angle B)$ or, $\angle XAY = \frac{1}{2}(180^\circ - \angle B - \angle C - 180^\circ + 2\angle B)$ $\therefore \angle XAY = \frac{1}{2}(\angle B - \angle C)$	5.	Subtracting (4) from (3)

Proved

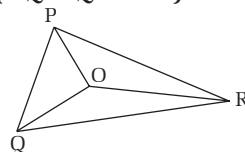
15. In $\triangle PQR$, O is the interior point. Prove that $OP + OQ + OR > \frac{1}{2}(PQ + QR + PR)$

Solution:

Given: In $\triangle PQR$, O is the interior point.

To prove: $OP + OQ + OR > \frac{1}{2}(PQ + QR + PR)$

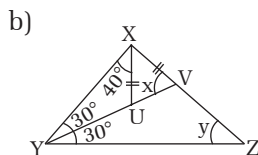
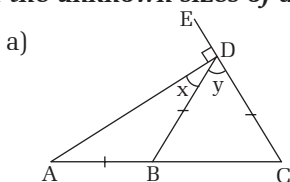
Proof:



	Statements		Reasons
1.	In $\triangle POQ$, $OP + OQ > PQ$	1.	The sum of any two sides of a triangle is greater than the third side
2.	In $\triangle QOR$, $OQ + OR > QR$	2.	Same as (1)
3.	In $\triangle POR$, $OP + OR > PR$	3.	Same as above
4.	$OP + OQ + OQ + OR + OP + OR > PQ + QR + PR$ or, $2OP + 2OQ + 2OR > PQ + QR + PR$ or, $2(OP + OQ + OR) > PQ + QR + PR$ $\therefore OP + OQ + OR > \frac{1}{2}(PQ + QR + PR)$	4.	Adding (1), (2) and (3)

Proved

16. Find the unknown sizes of angles in the following figures.



Solution:

a)

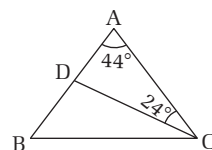
- i) In $\triangle ABD$, $\angle DAB = \angle ADB = x$ [$\because AB = BD$]
 ii) $\angle DBC = \angle DAB + \angle ADB = x + x = 2x$ [Being ext. angle of $\triangle$ equal to sum of opposite interior angles]
 iii. In $\triangle BCD$, $\angle DBC = \angle BCD = 2x$ [$\because BD = CD$]
 iv. In $\triangle ACD$, $\angle ACD + \angle CAD = \angle ADE$
 or, $x + 2x = 90^\circ \quad \therefore x = 30^\circ$
 v) In $\triangle BCD$, $\angle BCD + \angle CBD + \angle BDC = 180^\circ$
 or, $2x + 2x + y = 180^\circ$
 or, $4 \times 30 + y = 180^\circ$
 $\therefore y = 60^\circ$
 Hence, $x = 30^\circ, y = 60^\circ$

b)

- i) In $\triangle XYU$, $\angle XUV = \angle UXY + \angle UYX$ [Being ext. angle of $\triangle$ equal to sum of opposite interior angles]
 $= 40^\circ + 30^\circ$
 $= 70^\circ$
 ii) In $\triangle XUY$; $\angle XVU = \angle XUV$ [$\because XU = XV$]
 $\therefore x = 70^\circ$
 iii) In $\triangle VYZ$, $\angle VYZ + \angle YZV = \angle YVX$ [Being ext. angle of $\triangle$ equal to the sum of opposite angles.]
 or, $30^\circ + y = x$
 or, $30^\circ + y = 70^\circ \quad \therefore y = 40^\circ$

Hence, $x = 70^\circ$ and $y = 40^\circ$

17. In the given figure, $AB = AC$, $\angle BAC = 44^\circ$ and $\angle ACD = 24^\circ$, show that $BC = CD$.



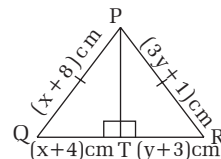
Solution:

- i) In $\triangle ABC$; $\angle BAC + \angle ABC + \angle ACB = 180^\circ$
 or, $44^\circ + \angle ABC + \angle ABC = 180^\circ$ [$\because \angle ABC = \angle ACB$]
 $\therefore \angle ABC = 68^\circ$
 ii) In $\triangle ACD$, $\angle CDB = \angle DAC + \angle ACD = 44^\circ + 24^\circ = 68^\circ$
 Since, $\angle ABC = \angle CDB = 68^\circ$.
 Hence, $BC = CD$

18. In the adjoining figure, find the value of x and y

Solution:

- i) $PQ = PR$ or, $x + 8 = 3y + 1 \quad \therefore x = 3y - 7$ (i)
 ii) $QT = TR$ [$\because PQ = PR$ and $PT \perp QR$]
 or, $x + 4 = y + 3$
 or, $x = y - 1$ (ii)
 Substituting the value of x in equⁿ (i), we get
 $3y - 7 = y - 1$ or, $2y = 6 \quad \therefore y = 3$
 Substituting the value of y in equⁿ (i), we get
 $x = 3 \times 3 - 7 = 2$
 Hence, $x = 2$ and $y = 3$



19. In the given figure, $AB = AC$, $BD = EC$ and $\angle DAE = 30^\circ$. Prove that $\triangle ADE$ is an isosceles triangle. Also calculate the size of $\angle ADE$.

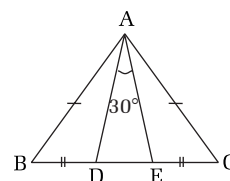
Solution:

Given: $AB = AC$, $BD = EC$ and $\angle DAE = 30^\circ$

To prove: $\triangle ADE$ is an isosceles triangle

To find: $\angle ADE$

Proof:



	Statements		Reasons
1.	In $\triangle ABD$ and $\triangle AEC$	1.	
i)	$AB = AC$ (S)	i)	Given
ii)	$\angle ABC = \angle ACB$ (A)	ii)	Base angles of an isosceles triangle are equal
iii)	$BD = EC$ (S)	iii)	Given
2.	$\triangle ABD = \triangle AEC$	2.	By S.A.S. axiom
3.	$AD = AE$	3.	Corresponding sides of congruent triangle are equal
4.	$\angle ADE = \angle AED$	4.	Base angles of isosceles triangle ADE
5.	In $\triangle ADE$; $\angle ADE + \angle AED + \angle DAE = 180^\circ$ or, $\angle ADE + \angle ADE + 30^\circ = 180^\circ$ $\therefore \angle ADE = 75^\circ$	5.	Sum of angles of triangle is 180° .

Proved

20. In the figure alongside, BO and CO are bisectors of $\angle ABC$ and $\angle ACB$ respectively. If $BO = CO$, prove that $\triangle ABC$ is an isosceles triangle.

Solution:

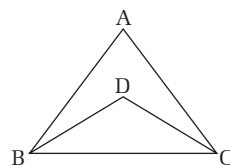
Given: i) BO and CO are the bisectors of $\angle ABC$ and $\angle ACB$ respectively.

$\therefore \angle ABC = 2\angle DBC$ and $\angle ACB = 2\angle DCB$.

ii) $BO = CO$

To prove: $\triangle ABC$ is an isosceles triangle

Proof:



	Statements		Reasons
1.	$\angle DBC = \angle DCB$	1.	$\because BO = CO$
2.	$\angle ABC = 2\angle DBC$ and $\angle ACB = 2\angle DCB$	2.	Given
3.	$\angle ABC = \angle ACB$	3.	From (1) and (2)
4.	$\triangle ABC$ is an isosceles triangle	4.	From (3)

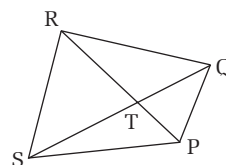
Proved

21. In the given figure PQRS, $SP = RQ$ and $RP = SQ$. Prove that $RT = ST$.

Solution:

Given: In the figure PQRS; $SP = RQ$ and $RP = SQ$

To prove: $RT = ST$



Proof:

	Statements		Reasons
1.	In $\triangle PRS$ and $\triangle QRS$	1.	
i.	$SP = RQ$ (S)	i.	Given
ii.	$RP = SQ$ (S)	ii.	Given
iii.	$RS = RS$ (S)	iii.	Side common to both $\triangle^s$
2.	$\triangle PRS \cong \triangle QRS$	2.	By S.S.S. axiom
3.	$\angle PRS = \angle QSR$	3.	Corresponding angles of congruent triangles are equal
4.	$\triangle RTS$ is an isosceles triangle	4.	From (3), base angles are equal
5.	$RT = ST$	5.	From (4)

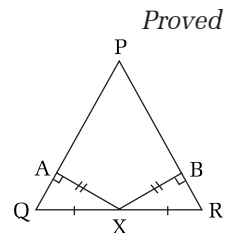
22. In the given figure, X is the mid - point of QR , $XA \perp PQ$, $XB \perp PR$ and $XA = XB$. Prove that $\triangle PQR$ is an isosceles triangle.

Solution:

Given: X is the mid - point of QR i.e, $QX = RX$, $XA \perp PQ$, $XB \perp PR$ and $XA = XB$.

To prove: $\triangle PQR$ is an isosceles triangle

Proof:



	Statements		Reasons
1.	In $\triangle AQX$ and $\triangle BRX$	1.	
i.	$\angle XAQ = \angle XBR$ (R)	i.	$XA \perp PQ$ and $XB \perp PR$
ii.	$QX = RX$ (H)	ii.	Given
iii.	$AX = BX$ (S)	iii.	Given
2.	$\triangle AQX \cong \triangle BRX$	2.	By R.H.S. axiom
3.	$\angle AQX = \angle BRX$ i.e, $\angle PQR = \angle PRQ$	3.	Corresponding angles of congruent triangles
4.	$\triangle PQR$ is an isosceles triangle	4.	From (3), base angles are equal

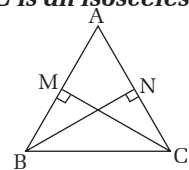
23. In the given figure $BN \perp AC$, $CM \perp AB$ and $BN = CM$. Prove that $\triangle ABC$ is an isosceles triangle.

Solution:

Given: $BN \perp AC$, $CM \perp AB$ and $BN = CM$

To prove: $\triangle ABC$ is an isosceles triangle.

Proof:



	Statements		Reasons
1.	In $\triangle BMC$ and $\triangle CNB$	1.	
i.	$\angle BMC = \angle CNB$ (R)	i.	Both are right angles
ii.	$BC = BC$ (H)	ii.	Common side
iii.	$CM = BN$ (S)	iii.	Given
2.	$\triangle BMC \cong \triangle CNB$	2.	By R.H.S axiom
3.	$\angle MBC = \angle NCB$ i.e, $\angle ABC = \angle ACB$	3.	Corresponding angles of congruent triangle are equal
4.	$\triangle ABC$ is an isosceles triangle	4.	From (3), being base angles equal

Proved

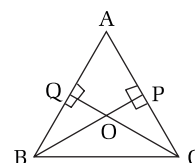
24. In the given triangle ABC , $AB = AC$, $BP \perp AC$ and $CQ \perp AB$.
Prove that (i) $BP = CQ$ (ii) $OP = OQ$

Solution:

Given: In $\triangle ABC$; $AB = AC$, $BP \perp AC$ and $CQ \perp AB$.

To prove: (i) $BP = CQ$ (ii) $OP = OQ$

Proof



	Statements		Reasons
1.	In $\triangle QBC$ and $\triangle PBC$	1.	
i.	$\angle BQC = \angle BPC$ (A)	i.	Both are right angles
ii.	$\angle QBC = \angle PCB$ (A)	ii.	Base angles of isosceles $\triangle ABC$
iii.	$BC = BC$ (S)	iii.	Common Side
2.	$\triangle QBC \cong \triangle PBC$	2.	By A.A.S axiom
3.	$CQ = BP$ i.e, $BP = CQ$	3.	Corresponding sides of congruent triangles
4.	$\angle BCQ = \angle PBC$	4.	Corresponding angles of congruent triangles
5.	$OB = OC$	5.	From (4), base angles of $\triangle OBC$ are equal
6.	$BP - OB = CQ - OC$ $\therefore OP = OQ$	6.	Subtracting (5) from (3)

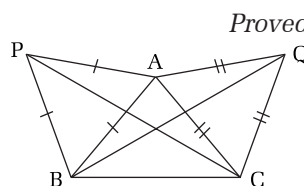
25. In the figure alongside, APB and AQC are equilateral triangles. Prove that: $PC = BQ$.

Solution:

Given: APB and AQC are equilateral triangles

To prove: $PC = BQ$

Proof



	Statements		Reasons
1.	$\angle PAB = \angle QAC = 60^\circ$	1.	Angles of equilateral triangles are equal
2.	$\angle PAB + \angle BAC = \angle QAC + \angle BAC$	2.	Adding $\angle BAC$ on both sides of (1)
3.	$\angle PAC = \angle QAB$	3.	By whole part axiom
4.	In $\triangle PAC$ and $\triangle QAB$	4.	
i.	$AP = AB$ (S)	i.	Sides of equilateral triangle APB
ii.	$\angle PAC = \angle QAB$ (A)	ii.	From (3)
iii.	$AC = AQ$ (S)	iii.	Sides of equilateral triangle AQC
5.	$\triangle PAC \cong \triangle QAB$	5.	By S.A.S. axiom
6.	$PC = BQ$	6.	Corresponding sides of congruent triangles

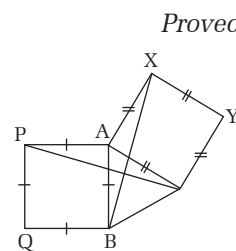
26. In the figure alongside $PABQ$ and CYX are squares.
Prove that: $PC = BX$

Solution:

Given: $PABQ$ and $ACYX$ are squares

To prove: $PC = BX$

Proof



	Statements		Reasons
1.	$\angle PAB = \angle CAX = 90^\circ$	1.	Angles of square are equal

2.	$\angle PAB + \angle BAC = \angle CAX + \angle BAC$	2.	Adding $\angle BAC$ on both sides of (1)
3.	$\angle PAC = \angle BAX$	3.	Whole part axiom
4.	In $\triangle PAC$ and $\triangle BAX$	4.	
i.	$PA = AB$ (S)	i.	Sides of square $PABQ$
ii.	$\angle PAC = \angle BAX$ (A)	ii.	From (3)
iii.	$AC = AX$ (S)	iii.	Being the sides of square $ACYX$
5.	$\triangle PAC \cong \triangle BAX$	5.	By S.A.S. axiom
6.	$PC = BX$	6.	Corresponding sides of congruent triangle are equal

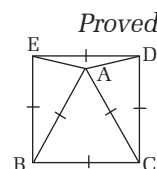
27. In the adjoining figure ABC is an equilateral triangle and $BCDE$ is a square. Prove that: $AE = AD$

Solution:

Given: ABC is an equilateral triangle and $BCDE$ is a square

To prove: $AE = AD$

Proof



	Statements		Reasons
1.	$\angle EBC = \angle BCD = 90^\circ$	1.	Angles of square are equal
2.	$\angle ABC = \angle BCA = 60^\circ$	2.	Angles of equilateral triangle
3.	$\angle EBC - \angle ABC = \angle BCD - \angle BCA$ $\therefore \angle EBA = \angle ACD$	3.	Subtracting (1) from (2)
4.	In $\triangle EBA$ and $\triangle ACD$	4.	
i.	$BE = CD$ (S)	i.	Sides of the square $BCDE$
ii.	$\angle EBA = \angle ACD$ (A)	ii.	From (3)
iii.	$AB = AC$ (S)	iii.	Sides of equilateral $\triangle ABC$
5.	$\triangle EBA \cong \triangle ACD$	5.	By S.A.S. axiom
6.	$AE = AD$	6.	Corresponding sides of congruent triangle are equal

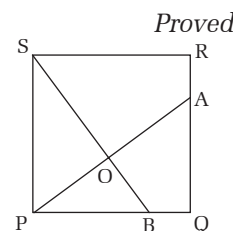
28. In the figure alongside, $PQRS$ is a square in which PA and SB intersect at O . If $PA = SB$, prove that PA and SP are perpendicular to each other at O

Solution:

Given: $PQRS$ is a square. $PA = SB$ and intersect at O .

To prove: $PA \perp SB$ at O . i.e, $\angle SOA = 90^\circ$

Proof



	Statements		Reasons
1.	In $\triangle SPB$ and $\triangle PAQ$	1.	
i.	$\angle SPB = \angle PQA$	i.	Both are right angles
ii.	$SB = PA$	ii.	Given
iii.	$SP = PQ$	iii.	Sides of square $PQRS$
2.	$\triangle SPB \cong \triangle PAQ$	2.	By R.H.S. axiom
3.	$\angle PSB = \angle APQ$	3.	Corresponding angles of congruent triangles

4.	$\angle SOA = \angle SPO + \angle PSO$	4.	Being ext. angle of triangle SOP equal to the sum of opposite interior angles
5.	$\angle SOA = \angle SPO + \angle OPB$	5.	From (3) and (4)
6.	$\angle SOA = \angle SPB = 90^\circ$	6.	From (5), by whole part axiom

29. In the figure alongside, $AB = AC$ and AD bisects $\angle CAE$. Prove that $AD \parallel BC$.

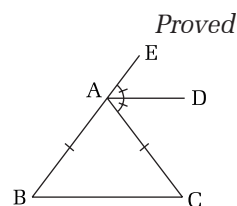
Solution:

Given: $AB = AC$ and AD bisects $\angle CAE$

To prove: $AD \parallel BC$

Proof:

	Statements		Reasons
1.	$\angle ABC = \angle ACB$	1.	$AB = AC$
2.	$\angle CAE = 2\angle CAD$	2.	$\angle CAD = \angle DAE$
3.	$\angle CAE = \angle ABC + \angle BCA$	3.	Being ext. angle of Δ equal to the sum of opposite interior angles
4.	$2\angle CAD = \angle BCA + \angle BCA$ $\therefore \angle CAD = \angle BCA$	4.	From (1), (2) and (3)
5.	$AD \parallel BC$	5.	From (4) alternate angles being equal



30. In the triangle given alongside, $PQ = PR$. The bisector of $\angle PQR$ meets PR at S . Prove that $\angle PSQ = 3\angle PQS$.

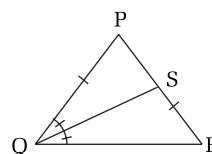
Solution:

Given: In $\triangle PQR$, $PQ = PR$, $\angle PQS = \angle SQR$

To prove: $\angle PSQ = 3\angle PQS$

Proof:

	Statements		Reasons
i.	$\angle PQR = \angle PRQ$	i.	$\therefore PQ = PR$
ii.	$\angle PQR = 2\angle PQS$	ii.	$\therefore \angle PQS = \angle SQR$
iii.	$\angle PRQ = 2\angle PQS$	iii.	From (i) and (ii)
iv.	In $\triangle QSR$; $\angle PSQ = \angle SQR + \angle SRQ$	iv.	
v.	$\angle PSQ = \angle PQS + 2\angle PQS$ $\therefore \angle PSQ = 3\angle PQS$	v.	From (iii) and (iv)



31. In the given figure, $BE = EC$ and CE is the bisector of $\angle ACB$. Prove that $\angle BEC = \angle ACD$.

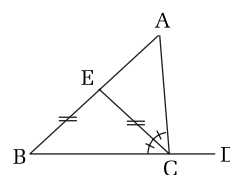
Solution:

Given: $BE = EC$ and $\angle ACE = \angle BCE$

To prove: $\angle BEC = \angle ACD$

Proof:

	Statements		Reasons
1.	$\angle EBC = \angle BCE$	1.	$BE = EC$



2.	$\angle BCE = \angle ACE$	2.	Given
3.	In $\triangle AEC$, $\angle BEC = \angle EAC + \angle ACE$	3.	Being ext. angle of $\triangle$ equal to the sum of opposite interior angles
4.	$\angle BEC = \angle BAC + \angle ABC$	4.	From (1), (2) and (3)
5.	In $\triangle ABC$, $\angle ACD = \angle BAC + \angle ABC$	5.	Same as (3)
6.	$\angle BEC = \angle ACD$	6.	From (4) and (5)

Proved

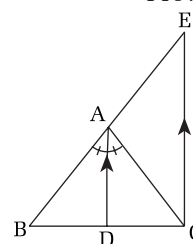
32. In the adjoining figure, AD is the bisector of $\angle BAC$ and $AD \parallel EC$. Prove that $AC = AE$.

Solution:

Given: $\angle BAD = \angle CAD$ and $AD \parallel EC$

To prove: $AC = AE$

Proof



	Statements		Reasons
1.	$\angle BAD = \angle CAD$	1.	Given
2.	$\angle BAD = \angle BEC$	2.	$AD \parallel EC$ and corresponding angles
3.	$\angle CAD = \angle ACE$	3.	$AD \parallel EC$ and alternate angles
4.	$\angle BEC = \angle ACE$	4.	From (i), (ii) and (iii)
5.	$AC = AE$	5.	From (iv), being base angles equal

Proved

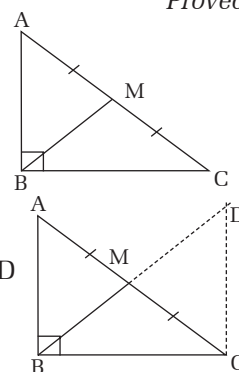
33. In $\triangle ABC$, $\angle ABC = 90^\circ$ and M is the mid-point of AC. Prove that: $AM = BM = CM$.

Solution:

Given: In $\triangle ABC$, $\angle ABC = 90^\circ$ and $AM = CM$

To prove: $AM = BM = CM$

Construction: BM is produced to D such that $BM = MD$ and CD is joined.



Proof

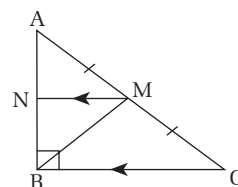
	Statements		Reasons
1.	In $\triangle ABM$ and $\triangle CDM$	1.	
i.	$AM = CM$ (S)	i.	Given
ii.	$\angle AMB = \angle CMD$ (A)	ii.	Vertically opposite angles are equal
iii.	$BM = MD$ (S)	iii.	By construction
2.	$\triangle ABM \cong \triangle CDM$	2.	By S.A.S. axiom
3.	$AB = CD$ and $\angle ABM = \angle CDM$	3.	Being corresponding sides and angles of congruent triangles
4.	$AB \parallel DC$	4.	Being alternate angles $\angle ABM$ and $\angle CDM$ equal from (3)
5.	$\angle ABC + \angle BCD = 180^\circ$ or, $90^\circ + \angle BCD = 180^\circ$	5.	$AB \parallel DC$ and co-interior angles $\therefore \angle BCD = 90^\circ$

6.	In $\triangle ABC$ and $\triangle BCD$,	6.	
i.	$AB = CD$ (S)	i.	From (3)
ii.	$\angle ABC = \angle BCD$ (A)	ii.	Both are right angles
iii.	$BC = BC$ (S)	iii.	Common side
7.	$\triangle ABC \cong \triangle BCD$	7.	By S.A.S axiom
8.	$\angle ACB = \angle DBC$	8.	Corresponding angles of congruent triangles are equal
9.	$BM = CM$	9.	From (8), being base angles of $\triangle MBC$ equal
10.	$AM = BM = CM$	10.	Given $AM = CM$ and from (9)

Alternative process: (Effective after teaching mid - point theorem)

Construction: $MN \parallel CB$ is drawn through M

Proof

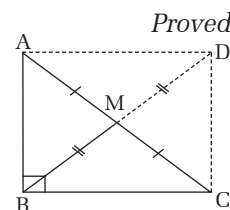


	Statements		Reasons
1.	$AN = BN$	1.	In $\triangle ABC$, $AM = CM$ and $MN \parallel CB$
2.	$\angle ANM = \angle ABC$	2.	$MN \parallel CB$, corresponding angles
3.	In $\triangle AMN$ and $\triangle BMN$	3.	
i.	$AN = BN$ (S)	i.	From (1)
ii.	$\angle ANM = \angle BNM$ (A)	ii.	From (2), both are right angles
iii.	$MN = MN$	iii.	Common side
4.	$\triangle AMN \cong \triangle BMN$	4.	By S.A.S axiom
5.	$AM = BM$	5.	Corresponding sides of congruent triangles are equal
6.	$AM = BM = CM$	6.	From (5) and given $AM = CM$

Alternative process:

Construction: BM is produced to D such that $BM = MD$ and quadrilateral $ABCD$ is completed.

Proof

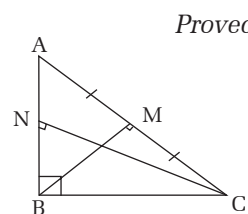


	Statements		Reasons
1.	$AM = CM$ and $BM = MD$	1.	Given and by construction
2.	$ABCD$ is a parallelogram	2.	From (1), being diagonals bisect each other
3.	$ABCD$ is a rectangle	3.	From (2) and being $\angle ABC = 90^\circ$
4.	$AC = BD$	4.	Diagonals of rectangle are equal
5.	$AM = CM = BM = MD$ $\therefore AM = BM = CM$	5.	From (1) and (4)

34. In the perpendiculars drawn from any two vertices to their opposite sides of a triangle are equal. Prove that the triangle is an isosceles triangle.

Solution:

Given: In $\triangle ABC$; $BM \perp AC$, $CN \perp AB$ and $BM = CN$



To prove: $\triangle ABC$ is an isosceles triangle

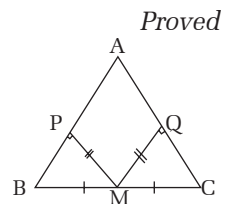
Proof

	Statements		Reasons
1.	In $\triangle NBC$ and $\triangle MBC$,	1.	
i.	$\angle BNC = \angle BMC$ (R)	i.	Both are right angles
ii.	$BC = BC$ (H)	ii.	Common sides
iii.	$CN = BM$ (S)	iii.	Given
2.	$\triangle NBC \cong \triangle MBC$	2.	By R.H.S. axiom
3.	$\angle NBC = \angle MCB$ i.e, $\angle ABC = \angle ACB$	3.	Corresponding angles of congruent triangle are equal
4.	$\triangle ABC$ is an isosceles triangle	4.	From (3), Both angles are equal

35. If the perpendiculars drawn from the mid - point of any side of a triangle to its other two sides are equal. Prove that the triangle is an isosceles triangle

Solution:

Given: In $\triangle ABC$, M is the mid point of side BC, $MP \perp AB$, $MQ \perp AC$ and $MP = MQ$



To prove: $\triangle ABC$ is an isosceles triangle

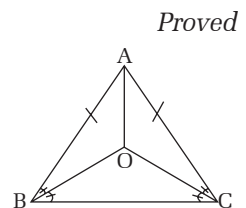
Proof

	Statements		Reasons
1.	In $\triangle PBM$ and $\triangle QMC$	1.	
i.	$\angle BPM = \angle CQM$ (R)	i.	Both are right angles
ii.	$BM = CM$ (H)	ii.	M being mid point of side BC
iii.	$MP = MQ$ (S)	iii.	Given
2.	$\triangle PBM \cong \triangle QMC$	2.	By R.H.S. axiom
3.	$\angle PBM = \angle QCM$ i.e, $\angle ABC = \angle ACB$	3.	Corresponding angles of congruent triangle
4.	$\triangle ABC$ is an isosceles triangle	4.	From (3), base angles are equal

36. Prove that the line joining the point of intersection of two angular bisectors of the base angles of an isosceles triangle to the vertex bisects the vertical angle.

Solution:

Given: In isosceles $\triangle ABC$, $AB = AC$, OB and OC are bisectors of base angles $\angle ABC$ and $\angle ACB$ respectively.



To prove: OA bisect $\angle BAC$ i.e, $\angle OAB = \angle OAC$

Proof

	Statements		Reasons
1.	$\angle ABC = \angle ACB$	1.	Base angles of isosceles triangle are equal
2.	$\angle ABC = 2\angle OBC$ and $\angle ACB = 2\angle OCB$	2.	Given
3.	$2\angle OBC = 2\angle OCB \therefore \angle OBC = \angle OCB$	3.	From (1) and (2)

4.	$OB = OC$	4.	Base angles of $\triangle OBC$ are equal
5.	In $\triangle AOB$ and $\triangle AOC$	5.	
i.	$AB = AC$ (S)	i.	Given
ii.	$OA = OA$ (S)	ii.	Common side
iii.	$OB = OC$ (S)	iii.	From (4)
6.	$\triangle AOB \cong \triangle AOC$	6.	By S.S.S. axiom
7.	$\angle OAB = \angle OAC$	7.	Corresponding sides of congruent triangles

37. In the angular bisector of an angle of triangle bisects the opposite side, prove that the triangle is an isosceles triangle.

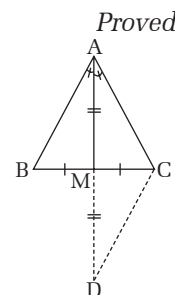
Solution:

Given: In $\triangle ABC$, $\angle BAM = \angle CAM$ and $BM = CM$

To prove: $\triangle ABC$ is an isosceles triangle i.e, $AB = AC$

Construction: AM is produced to D such that $AM = MD$ and CD is joined

Proof



	Statements		Reasons
1.	In $\triangle ABM$ and $\triangle CDM$	1.	
i.	$AM = MD$ (S)	i.	By construction
ii.	$\angle AMB = \angle CMD$ (A)	ii.	Vertically opposite angles
iii.	$BM = CM$ (S)	iii.	Given
2.	$\triangle ABM \cong \triangle CDM$	2.	By S.A.S. axiom
3.	$AB = CD$ and $\angle BAM = \angle CDM$	3.	Corresponding part of congruent triangles
4.	$\angle BAM = \angle CAM$	4.	Given
5.	$\angle CDM = \angle CAM$	5.	From (3) and (4)
6.	$AC = CD$	6.	From (5), base angles of $\triangle ACD$ are equal
7.	$AB = AC$	7.	From (3) and (6)

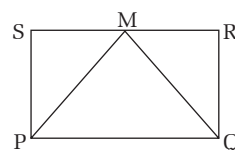
38. In the given rectangle PQRS, M is the mid - point of RS. Prove that PQM is an isosceles triangle.

Solution:

Given: In rectangle PQRS, M is the mid - point of RS.

To prove: PQM is an isosceles triangle

Proof



	Statements		Reasons
1.	In $\triangle PMS$ and $\triangle QRM$	1.	
i.	$SP = RQ$	i.	Opposite sides of rectangle are equal
ii.	$\angle PSM = \angle QRM$ (A)	ii.	Both are right angles
iii.	$SM = RM$ (S)	iii.	Given
2.	$\triangle PMS \cong \triangle QRM$	2.	By S.A.S. axiom
3.	$PM = QM$	3.	Corresponding sides of congruent triangle
4.	$\triangle PQM$ is an isosceles triangle	4.	From (3)

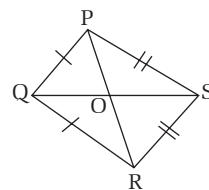
39. In the given figure, prove that QS is the perpendicular bisector of PR.

Solution:

Given: $PQ = QR$ and $PS = RS$

To prove: QS is perpendicular bisector of PR

Proof



	Statements		Reasons
1.	In $\triangle PQS$ and $\triangle QRS$	1.	
i.	$PQ = QR$ (S)	i.	Given
ii.	$PS = RS$ (S)	ii.	Given
iii.	$QS = QS$ (S)	iii.	Common side
2.	$\triangle PQS \cong \triangle QRS$	2.	By S.S.S. axiom
3.	$\angle PQS = \angle RQS$	3.	Corresponding angles of congruent triangle
4.	In $\triangle POQ$ and $\triangle ROQ$	4.	
i.	$PQ = QR$ (S)	i.	Given
ii.	$\angle PQO = \angle RQO$ (A)	ii.	From (3)
iii.	$QO = QO$	iii.	Common Side
5.	$\triangle POQ \cong \triangle ROQ$	5.	By S.A.S axiom
6.	$OP = OR$ and $\angle POQ = \angle ROQ$	6.	Corresponding parts of congruent triangles
7.	$\angle POQ = \angle ROQ = 90^\circ$	7.	Being adjacent angles on linear pair
8.	$OP = OR$ and $QS \perp PR$	8.	From (6) and (7)

Hence, QS is perpendicular bisector of PR.

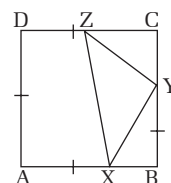
40. In the figure alongside, ABCD is a square. X, Y and Z are the points on the sides AB, BC and CD respectively. Such that $AX = BY = CZ$. Prove that XYZ is an isosceles triangle.

Solution:

Given: ABCD is a square. X, Y and Z are the point on the sides AB, BC and CD respectively.

To prove: XYZ is an isosceles triangle.

Proof



	Statements		Reasons
1.	$AB = BC$	1.	Adjacent sides of the square ABCD
2.	$AX = BY$	2.	Given
3.	$BX = CY$	3.	Subtracting (2) from (1)
4.	In $\triangle BXY$ and $\triangle CYZ$	4.	
i.	$BX = CY$ (S)	i.	From (3)
ii.	$\angle XBY = \angle YCZ$ (A)	ii.	Both are right angles
iii.	$BY = CZ$ (S)	iii.	Given
5.	$\triangle BXY \cong \triangle CYZ$	5.	By S.A.S axiom
6.	$XY = YZ$	6.	Corresponding sides of congruent triangles
7.	XYZ is an isosceles triangle	7.	From (6)

Proved

41. In the given figure, $PQ \parallel SR$. ST and RT bisect $\angle PSR$ and $\angle SRQ$ respectively. Prove that : $PQ = PS + QR$.

Solution:

Given: $PQ \parallel SR$. ST and RT bisect $\angle PSR$ and $\angle SRQ$ respectively. i.e., $\angle PST = \angle TSR$ and $\angle SRT = \angle QRT$

To prove: $PQ = PS + QR$

Proof

	Statements		Reasons
1.	$\angle PST = \angle TSR = \angle PTS$	1.	Given and alternate angles within $PQ \parallel SR$
2.	$PT = PS$	2.	From (1), $\angle PST = \angle PTS$
3.	$\angle QRT = \angle SRT = \angle QTR$	3.	Given and alternate angles within $PQ \parallel SR$
4.	$TQ = QR$	4.	From (3), $\angle QRT = \angle QTR$
5.	$PT + TQ = PS + QR$	5.	Adding (2) and (4)
6.	$PQ = PS + QR$	6.	From (5), whole part axiom

Proved

42. In the given figure, $BA \perp AC$, $RQ \perp PQ$, $AB = QR$ and $BP = CR$. Prove that $AC = PQ$.

Solution:

Given: $BA \perp AC$, $RQ \perp PQ$, $AB = QR$ and $BP = CR$

To prove: $AC = PQ$

Proof

	Statements		Reasons
1.	$BP = CR$	1.	Given
2.	$BC = PR$	2.	Adding PC on both side of (1)
3.	In $\triangle ABC$ and $\triangle PQR$	3.	
i.	$\angle BAC = \angle PQR$ (R)	i.	Both are right angles
ii.	$BC = PR$ (H)	ii.	From (2)
iii.	$AB = QR$ (S)	iii.	Given
4.	$\triangle ABC \cong \triangle PQR$	4.	By R.H.S. axiom
5.	$AC = PQ$	5.	Corresponding sides of congruent triangle are equal

Proved

43. In the figure alongside, $\angle OAD = \angle ODA$ and $\angle OBC = \angle OCB$. Prove that $AB = DC$.

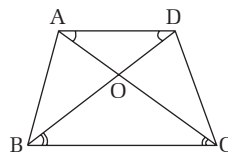
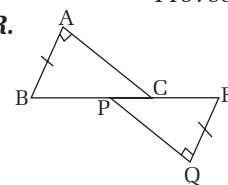
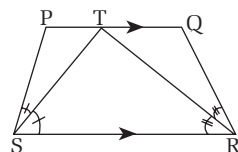
Solution:

Given: $\angle OAD = \angle ODA$ and $\angle OBC = \angle OCB$

To prove: $AB = DC$

Proof

	Statements		Reasons
1.	$OB = OC$	1.	$\therefore \angle OBC = \angle OCB$
2.	$OD = OA$	2.	$\therefore \angle ODA = \angle OAD$
3.	$BD = AC$	3.	Adding (1) and (2)



4.	In $\triangle DBC$ and $\triangle ABC$	4.	
i.	$BD = AC$ (S)	i.	From (3)
ii.	$\angle DBC = \angle ACB$ (A)	ii.	Given
iii.	$BC = BC$ (S)	iii.	Common side
5.	$\triangle DBC \cong \triangle ABC$	5.	By S.A.S. axiom
6.	$CD = AB$	6.	Corresponding sides of congruent triangle are equal

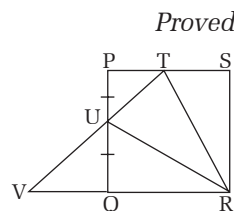
44. In the figure alongside, PQRS is a square. U is the mid - point of PQ, $\angle RUT = 90^\circ$, TU and RQ are produced to meet at V. Prove that : $TR = PT + PQ$

Solution:

Given: PQRS is a square, U is the mid - point of PQ and $\angle RUT = 90^\circ$

To prove: $TR = PT + PQ$

Proof



	Statements		Reasons
1.	In $\triangle PUT$ and $\triangle UVQ$	1.	
i.	$\angle TPU = \angle UQV$ (A)	i.	Both are right angles
ii.	$PU = UQ$ (S)	ii.	U is the mid - point of PQ
iii.	$\angle PUT = \angle VUQ$ (A)	iii.	Vertically opposite angles
2.	$\triangle PUT \cong \triangle UVQ$	2.	By A.S.A. axiom
3.	$PT = VQ$ and $UT = UV$	3.	Corresponding sides of congruent triangle are equal
4.	In $\triangle TUR$ and $\triangle UVR$	4.	
i.	$UT = UV$ (S)	i.	From (3)
ii.	$\angle TUR = \angle VUR$ (A)	ii.	Both are right angles
iii.	$UR = UR$ (S)	iii.	Common sides
5.	$\triangle TUR \cong \triangle UVR$	5.	By S.A.S. axiom
6.	$TR = VR$	6.	Corresponding sides of congruent triangles
7.	$VR = VQ + QR = PT + PQ$	7.	Whole part axiom, $VQ = PT$ and $QR = PQ$
8.	$TR = PT + PQ$	8.	From (6) and (7)

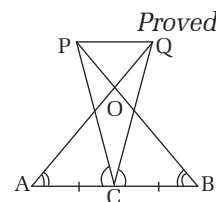
45. In the adjoining figure, $AC = BC$, $\angle PCA = \angle QCB$ and $\angle PBA = \angle QAB$. Prove that $\triangle OPQ$ is an isosceles triangle.

Solution:

Given: $AC = BC$, $\angle PCA = \angle QCB$ and $\angle PBA = \angle QAB$

To prove: $\triangle OPQ$ is an isosceles triangle

Proof



	Statements		Reasons
1.	$\angle PCA = \angle QCB$	1.	Given
2.	$\angle PCA + \angle PCQ = \angle QCB + \angle PCQ$	2.	Adding $\angle PCQ$ on both sides of (1)
3.	$\angle ACQ = \angle PCB$	3.	From (2), by whole part axiom

4.	In ΔQAC and ΔPCB	4.	
i.	$\angle QAC = \angle PBC$ (A)	i.	Given
ii.	$AC = BC$ (S)	ii.	Given
iii.	$\angle ACQ = \angle PCB$ (A)	iii.	From (3)
5.	$\Delta QAC \cong \Delta PCB$	5.	By A.S.A. axiom
6.	$\angle AQC = \angle BPC$	6.	Corresponding angles of congruent triangle
7.	$CQ = CP$	7.	Corresponding sides of congruent triangle
8.	$\angle CQP = \angle CPQ$	8.	From (7)
9.	$CQP - \angle AQC = \angle CPQ - \angle BPC$ i.e, $\angle OQP = \angle OPQ$	9.	Subtracting (6) from (8)
10.	ΔOPQ is an isosceles triangle	10.	Base angles are equal

Proved

Mid - point theorems

1. In ΔPQR , X and Y are mid - points of PQ and PR respectively. If $\angle P + \angle Q = 50^\circ$ and $\angle P + \angle R = 150^\circ$, find size of $\angle PXY$ and $\angle PYX$.

Solution:

In ΔPQR , X and Y are the mid - point of PQ and PR respectively. $\therefore$
 $XY \parallel QR$.

- i) $\angle R = 180^\circ - (\angle P + \angle Q) = 180^\circ - 50^\circ = 130^\circ$
 $\therefore \angle PYX = \angle R = 130^\circ$ [$\because XY \parallel QR$, corresponding angles]
 ii) $\angle Q = 180^\circ - (\angle P + \angle R) = 180^\circ - 150^\circ = 30^\circ$
 $\therefore \angle PXY = \angle Q = 30^\circ$ [$\because XY \parallel PQ$, corresponding angles]

2. In the given ΔABC , P, Q and R are the mid - point of sides AB, BC and CA respectively. If the perimeter of ΔPQR is 15 cm. find the perimeter of ΔABC .

Solution:

- i) $PQ = \frac{1}{2}AC$, $QR = \frac{1}{2}AB$ and $PR = \frac{1}{2}BC$

[line segment that joins the mid points of two sides is half of the third side in triangle.]

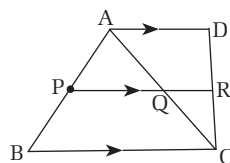
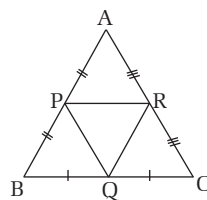
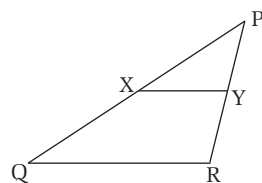
- ii) Perimeter of $\Delta PQR = 15$ cm
 or, $PQ + QR + PR = 15$ cm
 or, $\frac{1}{2}(AC + AB + BC) = 15$ cm
 $\therefore AB + BC + AC = 30$ cm

Hence, the perimeter of ΔABC is 30 cm.

3. In the given figures, $AD \parallel BC$ and P is the mid - point of AB. If $BC = 8$ cm and $QR = 3$ cm, find the length of PQ and AD.

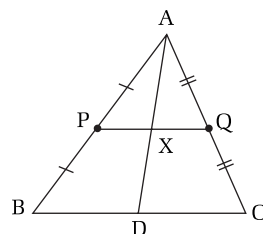
Solution:

- i) In ΔABC , $AQ = QC$ [$\because AP = PB$ and $PQ \parallel BC$]
 $PQ = \frac{1}{2}BC$ [PQ joins the mid - points of the sides AB and AC]
 $= \frac{1}{2} \times 8$ cm
 $= 4$ cm



- ii) In $\triangle ACD$; $CR = RD$ [$\therefore AQ = QC$ and $QR \parallel AD$]
 $\therefore QR = \frac{1}{2}AD$ [$\therefore Q$ and R joint the mid - point of AC and CD]
 or, $3 \text{ cm} = \frac{1}{2}AD$
 $\therefore AD = 6 \text{ cm}$

4. In the adjoining figures, P and Q are the mid - point of the sides AB and AC of $\triangle ABC$ respectively. X is a point on PQ . Prove that $AX = XD$.



Solution:

Given: In $\triangle ABC$, P and Q are the mid - point of the sides AB and AC respectively. X is a point on PQ .

To prove: $AX = XD$

Proof

	Statements		Reasons
1.	In $\triangle ABC$, $PQ \parallel BC$	1.	PQ joins the mid - points of the sides AB and AC
2.	In $\triangle ABD$; $AX = XD$	2.	$PX \parallel BD$ and P is the mid point of AB

Proved

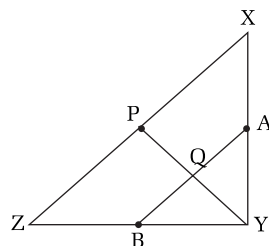
5. In the adjoining triangle XYZ , A and B are the mid - point of the sides XY and YZ respectively. P is any point on XZ . Prove that AB bisects PY at Q .

Solution:

Given: In $\triangle XYZ$; A and B are the mid - points of sides XY and YZ respectively. P is any point on XZ .

To prove: AB bisects PY at Q

Proof



	Statements		Reasons
1.	In $\triangle XYZ$, $AB \parallel XZ$	1.	AB joins the mid - points of XY and YZ
2.	In $\triangle PXY$, $YQ = QP$	2.	$AP \parallel XP$, and A is the mid - point of side XY
3.	AB bisects PY at Q	3.	From (2)

Proved

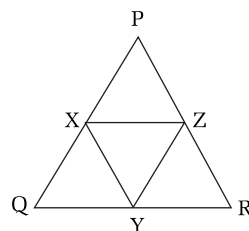
6. In the adjoining equilateral triangle PQR . X , Y and Z are the middle points of the sides PQ , QR and RP respectively. Prove that XYZ is also an equilateral triangle.

Solution:

Given: $\triangle PQR$ is an equilateral triangle. X , Y and Z are the mid - points of the sides PQ , QR and RP respectively.

To prove: $\triangle XYZ$ is also an equilateral triangle

Proof



	Statements		Reasons
1.	$PQ = QR = RP$	1.	$\triangle PQR$ is an equilateral triangle

2.	$YZ = \frac{1}{2}PQ$, $ZX = \frac{1}{2}QR$ and $XY = \frac{1}{2}RP$	2.	$\therefore$ Line joins the mid - points of any two sides of the triangle is half of the third side
3.	$YZ = ZX = XY$	3.	From (1) and (2)
4.	$\triangle XYZ$ is also the equilateral triangle	4.	From (3)

Proved

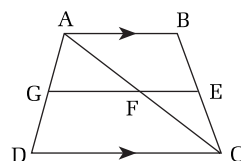
7. In the given figure, $AB \parallel DC$. If E is the mid - point of BC and F is the mid - point of AC , prove that G is the mid - point of AD .

Solution:

Given: $AB \parallel DC$, E is the mid - point of BC and F is the mid - point of AC .

To prove: G is the mid - point of AD .

Proof



	Statements		Reasons
1.	In $\triangle ABC$, $FE \parallel AB$	1.	EF join the mid - points of sides BC and AC
2.	$AB \parallel DC$	2.	Given
3.	$AB \parallel FE \parallel DC$ i.e., $AB \parallel GE \parallel DC$	3.	From (1) and (2)
4.	In $\triangle ADC$, G is the mid - point of AD	4.	$GE \parallel DC$ and F is the mid - point of AC

Proved

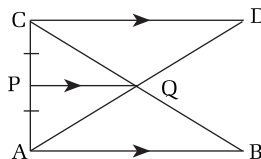
8. In the adjoining figure, $AB \parallel CD \parallel PQ$ and $AP = PC$. Prove that: $AB = CD$.

Solution:

Given: $AB \parallel CD \parallel PQ$ and $AP = PC$

To prove: $AB = CD$

Proof



	Statements		Reasons
1.	Q is mid - point of BC	1.	$\therefore$ In $\triangle ABC$, $AP = PC$ and $PQ \parallel AB$
2.	$PQ = \frac{1}{2}AB$	2.	PQ joins the mid - point of AC and BC in $\triangle ABC$
3.	Q is mid - point of AD	3.	$\therefore$ In $\triangle ADC$, $AP = PC$ and $PQ \parallel CD$
4.	$PQ = \frac{1}{2}CD$	4.	PQ joins the mid - point of AC and AD in $\triangle ADC$
5.	$AB = CD$	5.	From (ii) and (iv)

Proved

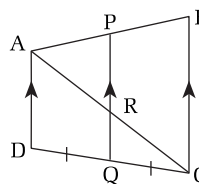
9. In the figure alongside, $AD \parallel PQ \parallel BC$ and $DQ = QC$. Prove that $AD + BC = 2PQ$.

Solution:

Given: $AD \parallel PQ \parallel BC$ and $DQ = QC$

To prove: $AD + BC = 2PQ$

Proof



	Statements		Reasons
1.	In $\triangle ADC$, $AR = RC$	1.	$DQ = QC$ and $RQ \parallel AD$

2.	$RQ = \frac{1}{2}AD \quad \therefore AD = 2RQ$	2.	RQ joins the mid - points of side AC and DC
3.	In $\triangle ABC$, $AP = PB$	3.	$AR = RC$ and $PR \parallel BC$
4.	$PR = \frac{1}{2}BC \quad \therefore BC = 2PR$	4.	$\therefore PR$ joins the mid - point of sides AB and AC
5.	$AD + BC = 2RQ + 2PR$	5.	Adding (ii) and (iv)
vi.	$AD + BC = 2PQ$	6.	Whole part axiom, $PR + RQ = PQ$

Proved

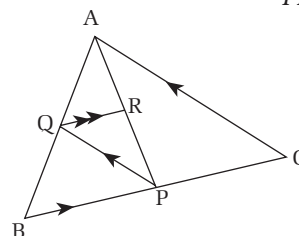
10. In the figure alongside, P is the mid - point of BC. PQ // CA and QR // BC. Prove that BC = 4QR.

Solution:

Given: P is mid - point of BC, PQ // CA and QR // BC

To prove: BC = 4QR

Proof



	Statements		Reasons
1.	In $\triangle ABC$, $AQ = BQ$	1.	$PQ \parallel CA$ and $BP = PC$
2.	In $\triangle ABD$, $AR = RP$	2.	$QR \parallel BC$ and $AQ = BQ$, from (1)
3.	$QR = \frac{1}{2}BP$	3.	From (1) and (2)
4.	$BP = \frac{1}{2}BC$	4.	P is the mid - point of BC
5.	$QR = \frac{1}{2} \times \frac{1}{2} BC$ $\therefore BC = 4QR$	5.	From (3) and (4)

Proved

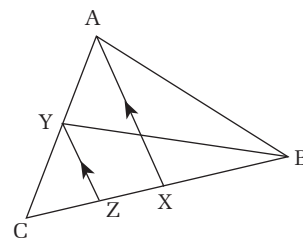
11. In the given $\triangle ABC$, AX and BY are medians, Z is a point on BC such that $YZ \parallel AX$. Prove that $BC = 4CZ$

Solution:

Given: In $\triangle ABC$, AX and BY are median. Z is a point on BC and $YZ \parallel AX$.

To prove: BC = 4CZ

Proof



	Statements		Reasons
1.	X is mid - point of BC and Y is the mid - point of AC	1.	Being AX and BY the median of $\triangle ABC$
2.	Z is mid - point of CX i.e, $CZ = ZX$ i.e, $CZ = \frac{1}{2}CX$ $\therefore CX = 2CZ$	2.	In $\triangle ACX$, $AY = YC$ and $YZ \parallel AX$
3.	$CX = \frac{1}{2}BC$	3.	$BX = CX$
4.	$2CZ = \frac{1}{2}BC$ $\therefore BC = 4CZ$	4.	From (2) and (3)

Proved

12. In the given figure, A is the mid - point of QR and B is the mid - point of PA. Prove that $PC = \frac{1}{3}PQ$.

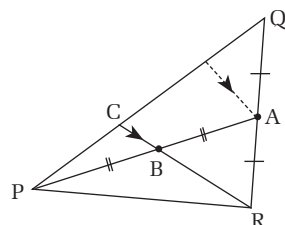
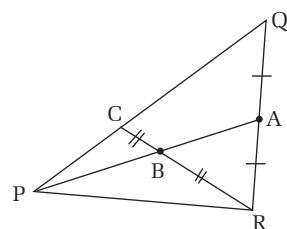
Solution:

Given: A is the mid - point of QR and B is the mid - point of PA

To prove: $PC = \frac{1}{3}PQ$

Construction: AD // BC is drawn where D is on PQ.

Proof:



	Statements		Reasons
1.	In $\triangle PAD$; $PC = CD$	1.	$PB = AB$ and $BC \parallel AD$
2.	In $\triangle QCR$, $CD = QD$	2.	$QA = AR$ and $AD \parallel CR$
3.	$PC = CD = QD$	3.	From (1) and (2)
4.	$PC = \frac{1}{3}PQ$	4.	From (3)

Proved

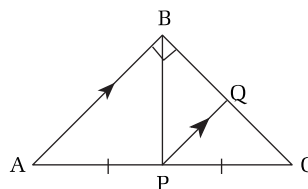
13. In the given right angled triangle ABC, right angled at B, P is the mid - point of AC. Prove that $BP = \frac{1}{2}AC$.

Solution:

Given: ABC is a right angled triangle, right angled at B, P is the mid - point of AC.

To prove: $BP = \frac{1}{2}AC$.

Proof:



	Statements		Reasons
1.	$BQ = QC$	1.	In $\triangle ABC$, $PQ \parallel AB$ and $AP = CP$
2.	$\angle PQC = \angle ABC = 90^\circ$	2.	$PQ \parallel AB$, corresponding angles
3.	In $\triangle PQC$ and $\triangle PQB$	3.	
i.	$CQ = BQ$ (S)	i.	From (1)
ii.	$\angle PQC = \angle PQB$ (A)	ii.	Both are right angles, From (2)
iii.	$PQ = PQ$ (S)	iii.	Common side
4.	$\triangle PQC \cong \triangle PQB$	4.	By S.A.S axiom
5.	$PC = BP$	5.	Corresponding sides of congruent triangles
6.	$PC = \frac{1}{2}AC$	6.	$\therefore AP = PC$
7.	$BP = \frac{1}{2}AC$	7.	From (5) and 6

Proved

14. In the given $\triangle ABC$, AP is the bisector of $\angle BAC$. If $BO \perp AP$ and $OQ \parallel AC$, prove that $BQ = QC$

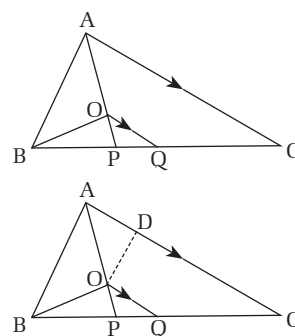
Solution:

Given: In $\triangle ABC$, AP is the bisector of $\angle BAC$. $BO \perp AP$ and $OQ \parallel AC$.

To prove: $BQ = QC$.

Construction: BO is produced to meet AC at D .

Proof:



	Statements		Reasons
1.	In $\triangle AOB$ and $\triangle AOD$	1.	
i.	$\angle BAO = \angle OAD$ (A)	i.	Given
ii.	$AO = AO$ (S)	ii.	Common side
iii.	$\angle AOB = \angle AOD$	iii.	Both are right angles
2.	$\triangle AOB \cong \triangle AOD$	2.	By A.S.A. axiom
3.	$BO = OD$	3.	Corresponding side of congruent triangle are equal
4.	$BQ = QC$	4.	In $\triangle BCD$, $BO = OD$ and $OQ \parallel DC$.

Proved

**15. In the given trapezium $PQRS$, A and B are the mid - point of the diagonals QS and PR respectively. Prove that (i) $AB \parallel SR$
(ii) $AB = \frac{1}{2}(SR - PQ)$**

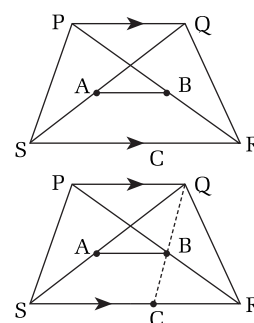
Solution:

Given: In trapezium $PQRS$, A and B are the mid - point of the diagonals QS and PR respectively.

To prove: (i) $AB \parallel SR$ (ii) $AB = \frac{1}{2}(SR - PQ)$

Construction: QB is produced to meet SR at C .

Proof:



	Statements		Reasons
1.	In $\triangle PQB$ and $\triangle CBR$	1.	
i.	$\angle QPB = \angle BRC$ (A)	i.	$PQ \parallel CR$ and alternate angles
ii.	$PB = BR$ (S)	ii.	B is the mid - point of PR
iii.	$\angle PBQ = \angle CBR$ (A)	iii.	Vertically opposite angles are equal
2.	$\triangle PQB \cong \triangle CBR$	2.	By A.S.A. axiom
3.	$BQ = BC$ and $PQ = CR$	3.	Corresponding sides of congruent triangles are equal
4.	$AB \parallel SC$ i.e, $AB \parallel SR$ and $AB = \frac{1}{2} SC$	4.	In $\triangle QSC$, AB joins the mid - point of sides QS and QC
5.	$SC = SR - CR$	5.	Subtraction axiom
6.	$AB = \frac{1}{2}(SR - PQ)$	6.	From (3), (4) and (5)

Proved

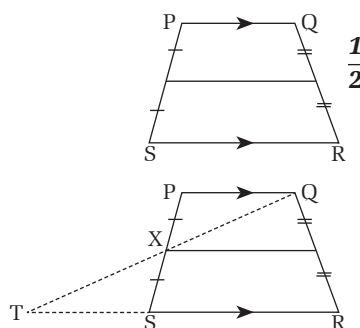
16. In the adjoining trapezium PQRS, X and Y are the mid - point of PS and QR respectively. Prove that: $XY = \frac{1}{2}(PQ + SR)$

Solution:

Given: X and Y are the mid - point of PS and QR respectively in trapezium PQRS.

To prove: $XY = \frac{1}{2}(PQ + SR)$

Construction: QX is joined and produced to meet RS produced at T.



Proof:

	Statements		Reasons
1.	In $\triangle PXQ$ and $\triangle TXS$	1.	
i.	$\angle XPQ = \angle TSX$ (A)	i.	$PQ \parallel TR$ and alternate angles
ii.	$PX = SX$ (S)	ii.	Given
iii.	$\angle PXQ = \angle TXS$ (A)	iii.	Vertically opposite angles are equal
2.	$\triangle PXQ \cong \triangle TXS$	2.	By A.S.A. axiom
3.	$PQ = TS$ and $QX = TX$	3.	Corresponding sides of congruent triangle
4.	$XY \parallel TR$ and $XY = \frac{1}{2} TR$	4.	XY join the mid - points of sides QT and QR respectively in $\triangle QTR$
5.	$TR = TS + SR$	5.	Whole part axiom
6.	$XY = \frac{1}{2}(PQ + SR)$	6.	From (3), (4) and (5)

Proved

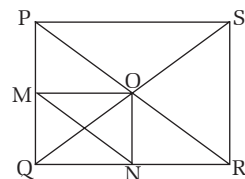
17. In the adjoining figure, PQRS and MQNO are rectangles. Prove that:
 (i) $MQ = \frac{1}{2} SR$ (ii) $MN = \frac{1}{2} QS$.

Solution:

Given: PQRS and MQNO are rectangles

To prove: (i) $MQ = \frac{1}{2} SR$ (ii) $MN = \frac{1}{2} QS$

Proof:



	Statements		Reasons
1.	$ON \parallel SR$	1.	$\therefore \angle ONQ = \angle SRQ = 90^\circ$, corresponding angles
2.	O is mid - point of QS	2.	Diagonals of rectangle bisect each other
3.	N is mid - point of QR	3.	From (1) and (2)
4.	$ON = \frac{1}{2} SR$	4.	From (2) and (3)
5.	$MQ = \frac{1}{2} SR$	5.	$ON = MQ$
6.	M is the mid - point of PQ	6.	$MO \parallel PS$ and $QO = OS$
7.	$MN \parallel PR$ and $MN = \frac{1}{2} PR$	7.	From (3) and (6)
8.	$MN = \frac{1}{2} QS$	8.	$\therefore PR = QS$, diagonals of rectangles

Proved

18. In the figure alongside, M is the mid - point of BC, Q is the mid - point of MR and $AB \parallel NM \parallel CQ$.

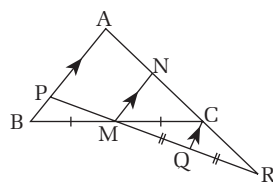
Prove that: (i) $PR = 3PM$ (ii) $AB = 4CQ$

Solution:

Given: M is the mid - point of BC, Q is the mid - point of MR and $AB \parallel NM \parallel CQ$.

To prove: (i) $PR = 3PM$ (ii) $AB = 4CQ$

Proof:



	Statements		Reasons
1.	$AN = NC$	1.	In $\triangle ABC$, $AB \parallel NM$ and $BM = MC$
2.	$MN = \frac{1}{2} AB$	2.	MN joins the mid - point of BC and AC respectively
3.	$NC = CR$	3.	In $\triangle NMR$, $NM \parallel CQ$ and $MQ = QR$
4.	$CQ = \frac{1}{2} NM$	4.	CQ joins the mid - point NR and MR respectively
5.	$CQ = \frac{1}{2} \times \frac{1}{2} AB$ $\therefore AB = 4CQ$	5.	From (2) and (4)
6.	In $\triangle PBM$ and $\triangle CQM$	6.	
i.	$\angle PBM = \angle MCQ$ (A)	i.	$PB \parallel CQ$ and alternate angles
ii.	$BM = MC$ (S)	ii.	Given
iii.	$\angle BMP = \angle CMQ$ (A)	iii.	Vertically opposite angles
7.	$\triangle PBM \cong \triangle CQM$	7.	By A.S.A. axiom
8.	$PM = MQ$	8.	Corresponding sides of congruent triangle are equal
9.	$PM = MQ = QR$	9.	Given and statement (8)
10.	$PR = 3PM$	10.	From (9)

Proved